Multiple Linear Regression

$$Y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3 + \dots$$

* How can we know which predictors are important?

In the book (Intro of Statistic Learning), the author gives an example, running predictor individually can be different from considering all predictors as multiple Linear regression:

page 72. table 3.3

	coeffics.	std.	t-stat.	p-value
simple regression on radio	0.203	0.020	9.92	<0.0001
~~coeffic~~ newspaper	0.055	0.017	3.30	<0.0001

↑

$y = \beta_0 + \beta_{radio} X_{radio}$, $y = \beta_0 + \beta_{newsp} X_{newspaper}$

table 3.4 multiple regression:

	coeffics.	std.	t-stat.	p-value
TV	0.046	0.0014	32.81	<0.0001
radio	0.189	0.0086	21.89	<0.0001
newspap	-0.001	0.0059	-0.18	0.8599

$y = \beta_0 + \beta_{TV} X_{TV} + \beta_{radio} X_{radio} + \beta_{newsp} X_{newsp}$

The coefficients have opposite sign!

→ counterintuitive eg. shark attack → ice cream post correlation page 75, table 3.5

correlation matrix

	TV	radio	news	sales
TV	1	0.054	0.0567	0.7822
radio		1	0.3541	0.5762
newspap			1	0.2283
sales				1

* the sales from newspaper could be due to correlation of newspaper and radio. But not necessarily mean newspaper influences to sales.

$\boxed{\text{Q1:}}$ any relation between the Response (y) and Predictors?

$H_0 = \beta_{TV} = \beta_{radio} = \beta_{newspap} = 0$

H_a : at least one $\beta_i \neq 0$

The hypothesis is given by F-statistic

$$\boxed{F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)}}$$

$TSS = \sum (y_i - \bar{y})^2$, $RSS = \sum (y_i - \hat{y}_i)^2$

(Table 3.6 of page 76)

In the book example, Regression result F-stat = 570 >> 1

meaning rejected H_0

(if $n >> p$) then $F > 1$ (slightly) still show evidence

if $n < p$) then need large F-stat to reject H_0

The p-value << 1 in the table 3.6, so we can reject H_0.

* if test partial set of Predictors.

$$H_0 = \beta_{p-q+1} = \beta_{p-q+2} = \cdots = \beta_p = 0 \quad , \quad q < p$$

f-statistics

$$\boxed{F\text{-stat} = \frac{(RSS_0 - RSS)/q}{RSS/(n-p-1)}}$$

Note pvalue in the table 3.4 shows TV & radio are related to Sales, but no evidence newspaper does.

But we need to do F-stat, not ~~t-stat~~, t-static only.

e.g. if $p=100$, $H_0: \beta_1 = \beta_2 = \cdots \beta_{100} = 0$

5% of p-value meaning. we still have $\beta_1 \cdots \beta_5$ could be wrong.

Col: if only rely on t-stat, we may make mistakes.

F-static needs to check.

Q2: Which Predictors are relevant? (if reject H_0)

We need to do Variable selection.

Mallows C_p, Akaike Info criterion, (AIC)

Bayes Info criterion (BIC)

adjusted R^2

* way to check:

① **forward selection:**

null model → fit

$$\left\{\begin{array}{l}\hat{y} = \theta_0 + \theta_1 x_1 \\ \hat{y} = \theta_0 + \theta_2 x_2 \leftarrow \text{lowest RSS} \\ \hat{y} = \theta_0 + \theta_3 x_3 \\ \vdots\end{array}\right.$$

$\rightarrow \hat{y} = \theta_0 + \theta_2 \hat{x}_2$

$$\rightarrow \left\{\begin{array}{l}\hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_2 \hat{x}_2 \\ \hat{y} = \theta_0 + \theta_3 \hat{x}_3 + \theta_2 \hat{x}_2 \\ \hat{y} = \theta_0 + \theta_4 \hat{x}_4 + \theta_2 \hat{x}_2 \\ \vdots \\ \hat{y} = \theta_0 + \theta_p \hat{x}_p + \theta_2 \hat{x}_2\end{array}\right. \longrightarrow \text{3 variables ..}$$

2 variables

② **backward selection:** (Doesn't work if $p > n$)

$$\hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_2 \hat{x}_2 + \cdots + \theta_p \hat{x}_p \longrightarrow \left\{\begin{array}{l}\theta_0 + \theta_1 \hat{x}_1 + \cdots + \hat{x}_{p-1}\theta_{p-1} \\ \theta_0 + \theta_1 \hat{x}_1 + \cdots + \hat{x}_p \theta_p \leftarrow \text{largest p-value} \\ \theta_0 + \theta_1 \hat{x}_1 + \theta_3 \hat{x}_3 + \cdots + \theta_p \hat{x}_p \\ \theta_0 + \quad \theta_2 \hat{x}_2 + \cdots + \theta_p \hat{x}_p\end{array}\right.$$

$\rightarrow \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \cdots + \theta_{p-2} \hat{x}_{p-2} + \theta_p \hat{x}_p$

$$\rightarrow \left\{\begin{array}{l}\hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \cdots + \theta_{p-2} \hat{x}_{p-2} \leftarrow \text{largest p-value} \Rightarrow \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \cdots + \theta_{p-3} \hat{x}_{p-3} + \theta_p \hat{x}_p \\ \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \cdots + \theta_p \hat{x}_p \\ \vdots \\ \hat{y} = \theta_0 + \quad \theta_2 \hat{x}_2 + \cdots + \theta_p \hat{x}_p\end{array}\right.$$

→ ...

③ **Mixed selection:**

null model $\rightarrow$ add one-by-one

$\hat{y} = \theta_0 + \theta_1 \hat{x}_1$ $\quad$ $\hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_2 \hat{x}_2 \leftarrow$

$\hat{y} = \theta_0 + \theta_2 \hat{x}_2$ $\quad$ - if pvalue large

$\vdots$ $\quad$ no consider $\hat{x}_2$

$\hat{y} = \theta_0 + \theta_p \hat{x}_p$

$\downarrow$

$$\begin{cases} \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_3 \hat{x}_3 \\ \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_4 \hat{x}_4 \\ \vdots \\ \hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_p \hat{x}_p \end{cases}$$

$\boxed{\text{Qualitative Predictors}}$

$$y = \beta_0 + \theta_1 x + \epsilon_i \quad \begin{cases} x=1. \text{ femal.} & \hat{y} = \theta_0 + \theta_1 + \epsilon \\ x=0 \text{ mal} & \hat{y} = \theta_0 + \epsilon \end{cases}$$

$$\text{Or } \hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \epsilon \begin{cases} \theta_0 + \theta_1 + \epsilon & \text{if } A \\ \theta_0 + \theta_2 + \epsilon & \text{if } B \\ \theta_0 + \epsilon & \text{else} \end{cases}$$

*Using dummy variables.

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extend ~~be~~ beyond linear Model

(a) <u>additive</u>: $\hat{Y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_2 x_2 + \epsilon$

the unit change in $\hat{x}_1$ or $\hat{x}_2$ is <u>independent</u> of $\hat{x}_2$ and $\hat{x}_1$.

→ interaction term:

$$\hat{y} = \theta_0 + \theta_1 \hat{x}_1 + \theta_2 \hat{x}_2 + \theta_3 \hat{x}_1 \hat{x}_2 + \epsilon$$

$$= \theta_0 + (\theta_1 + \theta_3 \hat{x}_2) \hat{x}_1 + \theta_2 \hat{x}_2 + \epsilon$$

$$= \theta_0 + \tilde{\theta}_1 \hat{x}_1 + \theta_2 \hat{x}_2 + \epsilon$$

now $\hat{\theta}_1 = \theta_1 + \theta_3 \hat{x}_2$, $\hat{x}_1$ is <u>no longer</u> indep of $\hat{x}_2$!

another eg. works and production lines:

$$y = \theta_0 + \theta_w X_{worker} + \theta_L X_{Lines} + \tilde{\theta} X_W X_L$$

∵ production lines need to have workers, otherwise nonsense.

→ $sales = \theta_0 + \theta_1 X_{TV} + \theta_2 X_{radio} + \theta_3 X_{radio} X_{TV} + \epsilon$

with interaction terms:

multiple reg:

		sd	t-stats	p-value		p-value
TV (θ_1)	0.0191	0.002	12.70	<0.0001	0.046	<0.0001
radio (θ_2)	0.0289	0.009	3.24	0.00014	0.189	<0.0001
TV×radio (θ_3)	0.0011	0.000	20.73	<0.0001 ←		

so θ_3 is <u>NOT</u> neglectable! p-value of TV×radio is low

→ $H_0: \theta_3 = 0$ can be rejected!